CURRICULUM VITAE

Matthew B. Jané

Forensic Meta-Scientist • PhD Candidate, Quantitative Psychology and Neuroscience

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Forensic meta-scientist who detects fabricated data, statistical errors, and research misconduct in published medical and clinical research. PhD candidate in Quantitative Psychology and Neuroscience at the University of Connecticut with expertise in meta-analysis and evidence synthesis. Creator of Audit Clinical Evidence (ACE), which audits published clinical trials for research integrity violations.

PROFESSIONAL EXPERIENCE

Forensic Analyst Aug 2025 – Dec 2025

The Medical Evidence Project, The Center for Scientific Integrity, New York, NY (remote)

- Investigated published medical and clinical research for fabricated or falsified data, statistical errors, and misconduct.
- Reanalyzed reported statistics and trial data to test internal consistency and plausibility; wrote investigation reports.

Graduate Assistant (Research and Teaching) Jan 2021 – Aug 2025

University of Connecticut, Storrs, CT

- Systematic Health Action Research Program (2022–2025; Dr. Blair Johnson): meta-analysis of health research.
- Sensory Perception and Neuroscience Laboratory (2020–2022; Drs. Read and Chrobak): experiments and data analysis.
- Teaching assistant: Principles of Psychological Research; Sensory Neuroscience Laboratory.

EDUCATION

Ph.D., Quantitative Psychology and Neuroscience Sep 2022 – Present

University of Connecticut · Advisor: Dr. Blair Johnson · GPA: 3.94

Dissertation: Correcting bias in meta-analysis: Comparing Rubin's effect size surface to Hunter and Schmidt's study-level correction procedures

M.S., Behavioral Neuroscience Jan 2021 – Aug 2022

University of Connecticut · Advisor: Dr. Heather Read · GPA: 3.97

B.S., Computational Neuroscience Aug 2016 – Dec 2020

University of Connecticut · Advisor: Dr. James Chrobak · GPA: 3.14

ADDITIONAL EXPERIENCE

Creator and Lead Auditor Sep 2026 – Present

Audit Clinical Evidence (ACE)

- Lead independent audits of published clinical research for statistical errors, data inconsistencies, and integrity violations.
- Direct study identification, statistical analyses, report development, and project standards.

Statistical Consultant Sep 2023 – May 2025

Self-employed

- Statistical consulting for labs and companies in genetics, agriculture, clinical psychology, and exercise physiology.

PUBLICATIONS

* Equal contribution.

Academic Publications

- Johnson, Simon, Li, Rokos, Martinez-Berman, **Jané**, Tsai, Loucks (2026) Mindfulness-based stress reduction interventions and cortisol regulation: A systematic review and meta-analysis. *Health Psychology*. Advance online publication. DOI: 10.1037/HEA0001636.
- **Jané***, Heathers*, Grimes* (2025) Major flaws in Taylor et al.'s (2025) meta-analysis on fluoride exposure and children's IQ scores. *OSF Preprints*. Under review at *Meta-Psychology*. DOI: 10.31219/OSF.IO/ZHM54_V3.
- **Jané**, Reinstein, Capitan (2025) Evaluation summary and metrics: "Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis". *The Unjournal*. DOI: 10.21428/D28E8E57.808D24C9.
- Johnson, **Jané**, Curley, Li, Martinez-Berman (2025) Methodological quality in health psychology research. *The Sage Handbook of Health Psychology*. Book chapter. DOI: 10.4135/9781529674743.N4.
- **Jané**, Harlow, Khu, Shah, Gould, Veiner, Kuo, Hsu, Johnson (2024) Extracting pre-post correlations for meta-analyses of repeated measures designs. Preprint. DOI: 10.17605/OSF.IO/HZJ4D.
- **Jané**, Xiao, Yeung, Ben-Shachar, Caldwell, Cousineau, Dunleavy, Elsherif, Johnson, Moreau, Riesthuis, Röseler, Steele, Vieira, Zloteanu, Feldman (2024) Guide to effect sizes and confidence intervals. Living open guide. DOI: 10.17605/OSF.IO/D8C4G.
- Harlow*, **Jané***, Read, Chrobak (2023) Memory retention following acoustic stimulation in slow-wave sleep: A meta-analytic review of replicability and measurement quality. *Frontiers in Sleep*. DOI: 10.3389/FRSLE.2023.1082253.
- **Jané**, Pisupati, Smith, Castro-Tonelli, Melo-Thomas, Schwarting, …, Read (2022) Correlations across timing cues in natural vocalizations predict biases in judging synthetic sound burst durations. *bioRxiv*. DOI: 10.1101/2022.05.29.493898.

Public Writing

- Meyerowitz-Katz, **Jané** (2024) Governments are pushing teen social media bans – but behind the scenes is a messy fight over science. *The Conversation*. DOI: 10.64628/AA.WAGGH46CA.

TALKS AND WORKSHOPS

Princeton University – *Detecting errors and fraud in meta-analysis with the help of forensic meta-science* 2025
Invited talk and workshop, Seminar in Advanced Research Methods

Physalia Courses – *Meta-analysis in R* 2025
Four-day workshop

University of South Florida – *Meta-analysis and data visualization in R* 2024
Two 1-hour workshops

R Workshops for Ukraine – *Meta-analysis in R* 2024
2-hour charity workshop

SOFTWARE

- **ThemePark** R PACKAGE Popular-culture styled ggplot themes, featured on R Weekly and FlowingData. github.com/MatthewBJane/theme_park
- **POSC** R PACKAGE Probability of Outcome Superiority Curves (POSCs). github.com/MatthewBJane/posc
- **OpenSynthesis** DATABASE A catalog of publicly available meta-analytic databases. matthewbjane.com/opensynthesis/
- **Interactive Simulators** WEB APPS Browser-based simulators for teaching and research on effect sizes and bias.
  - *Artifact Simulator*: How measurement error, range restriction, dichotomization, misclassification, and selection bias distort correlations, standardized mean differences, odds ratios, and risk ratios. matthewbjane.com/artifact-simulator.html
  - *Clinical Trial Simulator*: How non-adherence, attrition, measurement error, unblinded assessment, and failed allocation concealment bias randomized trial estimates. matthewbjane.com/trial-simulator.html
  - *POSC Simulator*: Interactive probability of outcome superiority curves. matthewbjane.com/posc-simulator.html

MEDIA COVERAGE

- Meyerowitz-Katz (2026) How do we deal with seriously problematic research? *Medium*. On Audit Clinical Evidence (ACE), the clinical trial auditing project. gidmk.medium.com.
- Ritchie, Chivers (2025) Jonathan Haidt vs. social media. *Science Fictions (podcast)*. On the Haidt–Jané debate over the social media reduction meta-analysis. sciencefictionspod.substack.com.
- The research on fluoride and IQ (2025). *Good Science Project*. On the fluoride–IQ critique. goodscienceproject.org.
- Reinstein (2025) Can we do useful meta-analysis? Unjournal evaluations of "Meaningfully reducing consumption of meat and

animal products is an unsolved problem". *Effective Altruism Forum*. Discusses Jané's Unjournal evaluation. forum.effectivealtruism.org.

- Steier (2025) Fighting for the F word: Defending fluoride against fear and faulty science. *Unbiased Science*. On the fluoride–IQ critique. theunbiasedscipod.substack.com.
- Rausch, Haidt (2024) The fundamental flaws of the only meta-analysis of social media reduction experiments (and why it matters), part 2. *After Babel*. Response to Jané's critique of their re-analysis. afterbabel.com.
- Lee (2024) Jonathan Haidt started a social-media war. Did he win? *The Chronicle of Higher Education*. On the Haidt–Jané social media debate. chronicle.com.

SKILLS AND PROFESSIONAL SERVICE

Statistical methods	Meta-analysis, forensic meta-science, clinical trials, regression, simulation.
Research integrity	Detecting statistical errors, data inconsistencies, and fabricated or manipulated data.
Programming and tools	R, Python, MATLAB, Quarto, LaTeX, Git and GitHub.
Science communication	Public writing, social media, workshops, and invited talks.
Peer review	*Psychological Bulletin*, *Collabra: Psychology*, *British Journal of Psychology*, *The Unjournal*.
Memberships	Science Integrity Alliance.